

Kiwix English Best — User Guide

Version: v20260822b

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Overview

Kiwix English Best is a Python script that automates the selection, downloading, and maintenance of the best available English-language ZIM files from the Kiwix download mirrors. ZIM files are compressed offline archives used by the [Kiwix](#) reader to browse Wikipedia, Stack Exchange, Project Gutenberg, FreeCodeCamp, and hundreds of other resources without an internet connection.

The script:

- Crawls the Kiwix server to find the best (newest, highest-quality) English ZIM file for each content group
- Downloads files using BitTorrent (via libtorrent) where available, with HTTP as a fallback
- Supports resuming interrupted downloads
- Verifies file integrity after download

- Detects and downloads updates when newer versions are available on the server
 - **Deletes outdated versions immediately** after each replacement is verified (one file at a time, not at session end)
 - **Accounts for reclaimable disk space** when calculating required free space (subtracts size of old versions that will be replaced)
 - Reports all sizes in binary units (KiB / MiB / GiB / TiB) with two decimal places
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Requirements

Requirement	Notes
Python 3.8+	Required
requests	<code>pip install requests</code>
tqdm	<code>pip install tqdm</code> (optional, for progress bars)
libtorrent	Optional; enables torrent downloads
zimcheck	Optional; enables deep ZIM integrity checking (HTTP downloads only)

Install Python dependencies:

```
pip install requests tqdm
```

To enable torrent support, install libtorrent for your platform. On Debian/Ubuntu:

```
sudo apt install python3-libtorrent
```

Getting Started

Run the script interactively:

```
python3 kiwix_english_best_v20260822b.py
```

You will see the main menu:

```
Kiwix English ZIM Tool v20260822b
```

- 1) Generate / download new best English ZIMs list
 - 2) Download / resume from existing best English ZIMs list
 - 3) Check for updates (new and newer ZIMs; includes cleanup)
 - 4) Cleanup ZIMs directory (remove older duplicate versions)
- Choose (1/2/3/4):

All ZIM files are stored in the `zims/` subdirectory relative to the script's location.

Menu Options

Option 1 — Generate / Download New Best English ZIMs List

When to use: First-time setup, or when you want to refresh the full best-file selection from the server.

What it does:

1. Crawls the Kiwix server to discover all available English ZIM files
2. Selects the best (newest date, then largest / highest priority) file per content group, applying the exclusion rules below
3. Saves the complete best list to `kiwix_english_best.txt` — this always happens, even if all files are already present on disk
4. Checks which files from the list are not yet downloaded (tolerant size match)
5. If files are missing, shows a space check (including reclaimable space from any older same-group files) and prompts to download
6. After each successful download, immediately deletes any older version in the same group

Prompts:

Prompt	Description
Download now? (y/n)	Whether to start downloading the missing files
Verbose output? (y/n)	Verbose shows per-file progress lines; non-verbose shows a compact updating display

Note: The list file (`kiwix_english_best.txt`) is always saved with the full best selection, regardless of what is already on disk. This file is used by Option 2 for resuming downloads.

Option 2 — Download / Resume from Existing List

When to use: Resume a previously interrupted download session, or re-download missing files from an existing list.

What it does:

1. Checks whether `kiwix_resume_update.txt` exists (left by a failed Option 3 session) — if so, offers to resume those incomplete update downloads first
2. Otherwise loads `kiwix_english_best.txt`
3. Filters out files already present on disk (using a size-tolerance check: max of 50 MiB or 5% of expected size)
4. Shows a space check for remaining files (gross size, reclaimable from old versions, net + buffer)
5. Downloads any incomplete or missing files, resuming `.partial` files where they exist

6. After each successful download, immediately deletes any older version in the same group

Prompts:

Prompt	Description
Resume these downloads? (y/n)	Shown only if <code>kiwix_resume_update.txt</code> exists
Verbose output? (y/n)	Progress display mode

Tip: If a download was interrupted, simply re-run Option 2. The script automatically detects .partial files and resumes from where it left off.

Option 3 — Check for Updates

When to use: Periodically, to download newer versions of ZIM files and pick up newly available content groups.

What it does:

1. **Checks for an incomplete previous update session** — if `kiwix_resume_update.txt` exists, offers to resume it (skipping the server crawl entirely), or to run a fresh check
2. **Crawls the server** and compares the best available version of each group against what is on disk (using content date from filenames)
3. **Reports** the number of updates (newer versions of files you already have) and new groups (content you don't have yet), with sizes
4. **Checks for torrent availability** for each file to download
5. **Shows a space check** that subtracts reclaimable space from outdated files that will be replaced
6. **Downloads** updates and new groups; **deletes each outdated predecessor immediately** after its replacement is verified (one file at a time, not at session end)
7. **Saves a resume file** (`kiwix_resume_update.txt`) if any downloads fail, so they can be resumed by running Option 3 again

Prompts:

Prompt	Description
r) Resume / n) Check for all updates	Only shown if <code>kiwix_resume_update.txt</code> exists
Download updates? (y/n)	Whether to start downloading
Verbose output? (y/n)	Progress display mode

Important notes:

- Option 3 uses **torrent downloads** where available, falling back to HTTP

- Cleanup of the old version happens **immediately** after each file is verified — so if the session is interrupted, already-completed updates have already freed their old space
 - If any downloads fail, `kiwix_resume_update.txt` is written automatically. Next time you run Option 3, you'll be offered the choice to resume it
-

Option 4 — Cleanup ZIMs Directory

When to use: If you have accumulated older duplicate versions of ZIM files (e.g. after manually copying files without using the script's auto-cleanup).

What it does:

1. Scans `zims/` for files belonging to the same content group
2. For each group with multiple versions, shows which file will be **kept** (newest date) and which will be **deleted**
3. Shows the total space that will be freed
4. Asks for confirmation before deleting anything

Confirmation prompt:

Proceed with deletion? (y/n):

Answering n cancels without touching any files.

Example output:

Duplicate groups found: 2

```
Group 'wikipedia_en_all_maxi':
  Keep    : wikipedia_en_all_maxi_2026-07.zim
  Delete  : wikipedia_en_all_maxi_2026-04.zim (115.00 GiB)
```

```
Group 'devdocs_en_axios':
  Keep    : devdocs_en_axios_2026-06.zim
  Delete  : devdocs_en_axios_2026-02.zim (330.00 KiB)
```

2 file(s) will be deleted, freeing 115.00 GiB.

Proceed with deletion? (y/n):

Selection Rules

The crawler keeps only the most comprehensive English variant in each content group. Priority is:

1. `_all_maxi_` (highest)
2. Other comprehensive `_all_` forms

3. Everything else (only if no better option exists)

Hard exclusions:

Rule	Effect
Wikipedia	Keep only wikipedia_en_all_maxi_*. Drop topic splits, _mini_, _nopic_, _simple_all_, _top_*, _wpl_*, etc.
Gutenberg	Keep only gutenberg_en_all_*. Drop all gutenberg_en_lcc-* letter splits.
Wiktionary	Keep wiktionary_en_all_nopic_* (the only comprehensive English option available).
FreeCodeCamp	Keep only freecodecamp_en_all_*. Drop all topic subset ZIMs.
Speedtest / diagnostics	Drop all speedtest_* files.
Other nopic	Drop most remaining _nopic_* variants.
Regional subsets	Drop wikivoyage_en_europe_* and similar partials.

Within a group, files are ranked by: selection priority – content date (newer first) – size (larger first) – filename.

Space Checks and Immediate Cleanup

Space check breakdown

Before every download session (Options 1, 2, 3, and cron mode), the script reports:

Line	Meaning
Download size (gross)	Bytes still to write for incomplete or missing files
Reclaimed from old versions	On-disk size of same-group files that will be deleted when replacements succeed (lists filenames)
Required (net + buffer)	$\max(0, \text{gross} - \text{reclaimed}) + \text{safety buffer}$ (default 10 GiB, or 10% of net, whichever is larger)
Available	Free space on the volume holding zims/

Example:

```
Space check for updates:
  Files to download: 3
```

```
Download size (gross): ~669.90 GiB
Reclaimed from old versions: ~612.00 GiB (2 file(s))
  - gutenbergr_en_all_2025-11.zim (206.00 GiB) → replaced by
gutenbergr_en_all_2026-03.zim
  - wikipedia_en_all_maxi_2026-01.zim (406.00 GiB) → replaced by
wikipedia_en_all_maxi_2026-07.zim
Required (net + 10 GiB buffer): ~67.90 GiB
Available: ~1.08 TiB
```

If there are no older versions to replace, the reclaim line shows none.

Immediate per-file cleanup

After **each** successful download and integrity verification — in any menu option — the script deletes older same-group .zim files **immediately**, without waiting for the rest of the batch:

```
Removed old version: gutenbergr_en_all_2025-11.zim (freed 206.00 GiB)
```

Concurrent downloads (default max 4) each free their own predecessor when they finish. Peak concurrent usage can temporarily exceed the *net* figure while several large files are in flight before their old counterparts are removed; the safety buffer and the warning text on the insufficient-space prompt reflect that.

Non-Interactive Update Mode (Cron Jobs)

The script supports a fully non-interactive update mode suitable for automated scheduling via cron or any task scheduler.

Usage

```
python3 kiwix_english_best_v20260822b.py -u
# or
python3 kiwix_english_best_v20260822b.py --update
```

Behaviour

- **Exits silently** (exit code 0) if the zims/ directory does not exist — this prevents cron from sending failure emails when the storage volume is not mounted (e.g. VeraCrypt volume not unlocked)
- If `kiwix_resume_update.txt` exists, resumes the incomplete session
- Otherwise crawls the server and downloads all updates and new groups
- Applies the same reclaimable-space calculation and immediate per-file cleanup as interactive mode
- Non-verbose output (clean for log files)
- Exits with code 1 if there is insufficient disk space or the server crawl fails

Example Cron Entry

To run every Sunday at 3:00 AM and log output:

```
0 3 * * 0 python3 /path/to/kiwix_english_best_v20260822b.py -u >>
/path/to/kiwix_cron.log 2>&1
```

Important: If your ZIM files are on an encrypted volume (e.g. VeraCrypt), the volume must already be mounted when the cron job runs. The script will exit gracefully if it is not.

File Reference

File	Description
zims/	Directory where all ZIM files are stored
zims/*.zim	Completed, verified ZIM files
zims/*.zim.partial	Incomplete downloads — safe to leave in place; will be resumed automatically
zims/*.zim.fastresume	Torrent resume data — deleted automatically on completion
zims/corrupt/	Files that failed integrity verification
kiwix_english_best.txt	Full best-file list generated by Option 1
kiwix_resume_update.txt	Incomplete update session resume file — written by Option 3, deleted on full completion
kiwix_download_failures.log	Log of all download failures with timestamps and error details

List File Format

kiwix_english_best.txt and kiwix_resume_update.txt use a pipe-delimited format:

```
# Format: filename|size_human|size_bytes|url|torrent_url
wikipedia_en_all_maxi_2026-07.zim|115.00 GiB|123456789012|https://...|
https://...torrent
```

Both old format (filename|size_bytes|url|torrent_url) and new format (filename|size_human|size_bytes|url|torrent_url) are accepted when loading a list.

Download Methods

Torrent (Preferred)

When a .torrent file is available on the Kiwix server for a ZIM file, the script downloads via BitTorrent using libtorrent. This is faster and more reliable for large files, as it distributes the load across multiple peers.

The progress display shows status, percentage, speed, and peer counts. Torrent downloads are **piece-verified** by libtorrent, so only a ZIM magic-header check is performed after completion (full zimcheck would be redundant and slow).

HTTP (Fallback)

If no torrent is available, or if the torrent download fails, the script falls back to direct HTTP download from the configured mirrors. Resume is supported via HTTP Range requests — interrupted downloads pick up from the byte offset already on disk.

Default concurrency is **4** parallel downloads. Space checks use the net-after-reclaim figure plus a safety buffer (default 10 GiB, or 10% of net, whichever is larger).

Progress Display

Verbose Mode

Each active download gets its own status line. A total progress bar tracks cumulative bytes downloaded versus the session total, with a byte-based ETA:

```
gutenberg_en_all_2026-03.zim - downloading | 12.4% | 5.2 MB/s | Peers: 11
khanacademy_en_all_2023-03.zim - downloading | 3.1% | 2.1 MB/s | Peers: 8
Total progress:  4%|| | 24.5G/609G [00:12:41, 780MB/s, ETA=01:15:22]
```

When an update finishes, you may also see:

```
Removed old version: gutenberg_en_all_2025-11.zim (freed 206.00 GiB)
```

Non-Verbose Mode

A single updating block shows active files plus the total bar, rewritten in place without scrolling.

Post-Run Summary

After every download session, a summary is printed with good / corrupt counts and total disk usage under `./zims/`. Incomplete files (if any) are listed with bytes on disk and bytes remaining, plus a pointer to the appropriate resume option.

Integrity Verification

After every successful download, the file is verified before being counted as complete (and before any old version is deleted):

Download Method	Verification
Torrent	libtorrent piece-verification (all pieces checked during download); ZIM magic header check only after completion
HTTP	ZIM magic header check; <code>zimcheck -C -D</code> (structure and directory integrity) if <code>zimcheck</code> is installed

Note: The `-I` flag (index check) is intentionally omitted from `zimcheck` for performance. Index integrity can be checked manually if needed:

```
zimcheck -I yourfile.zim
```

Files that fail verification are moved to `zims/corrupt/` and logged to `kiwix_download_failures.log`. Failed downloads do **not** trigger deletion of the previous good version.

Server Layout Note (August 2026)

As of August 2026, <https://download.kiwix.org/zim/> permanently redirects to the Kiwix Hub marketing site (<https://hub.kiwix.org/downloads/>), which does **not** provide a classic Apache directory listing.

This script therefore crawls:

```
https://lb.download.kiwix.org/zim/
```

which still serves the traditional index of category directories (`wikipedia/`, `gutenberg/`, `stack_exchange/`, `zimit/`, etc.). Individual file download URLs on `download.kiwix.org` continue to work.

If a future layout change removes the Apache `<pre>` listing from the configured base URL, the script prints:

```
WARNING: No Apache <pre> listing found at <url>
The site layout may have changed. Check BASE_URL / mirrors.
```

and returns an empty result set instead of failing silently.

Tips and Troubleshooting

A download failed — how do I resume it?

- If the failure happened during an **Option 3** update session, re-run **Option 3**. You will be offered the chance to resume from `kiwix_resume_update.txt`.
- Otherwise re-run **Option 2**. The script detects `.partial` files and resumes automatically.

- Already-completed replacements in that session will already have had their old versions deleted.

The server crawl returns zero files

1. Confirm internet connectivity.
2. Confirm the script version is **v20260822a or later** (older versions still pointed at `download.kiwix.org/zim/`, which no longer has a parseable listing).
3. Look for the WARNING: No Apache <pre> listing found... message — that indicates a further site-layout change; update `BASE_URL` / `MIRRORS` in the script.

The server crawl is very slow

The script scans each category subdirectory in sequence. This is normal — the server has many directories. A full scan typically takes one to a few minutes depending on network conditions.

I see “Found 0 update(s)” but I know there are new files

Check that your on-disk files have dates in their filenames (e.g. `_2026-07`). Option 3 compares content dates parsed from filenames. Files without a recognisable `YYYY-MM` date pattern will not be compared correctly.

Torrent downloads show 0 seeds / peers

Wait a minute or two — peers are discovered gradually. If activity remains at zero for several minutes, the torrent may be poorly seeded. The script will fall back to HTTP if the torrent path fails.

A file ended up in `zims/corrupt/`

The file failed integrity verification. The previous good version (if any) was **not** deleted. You can attempt to re-download by running Option 3 or by deleting the corrupt entry and running Option 2.

How do I see what files are scheduled for download?

Open `kiwix_english_best.txt` in any text editor. Each line lists a filename, its human-readable size, byte size, HTTP URL, and torrent URL (when available).

Space check says I need more room than expected

The required figure is **net** after reclaiming old versions, plus a safety buffer. Peak concurrent usage (up to 4 parallel downloads) can still briefly exceed the net figure while new files are writing and old ones have not yet been deleted. If the drive is truly tight, reduce concurrent downloads or free space manually before starting.

When are old files actually deleted?

Immediately after each replacement finishes downloading **and** passes integrity verification — one file at a time, as that worker completes. The script does **not** wait until the entire download batch is finished.

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